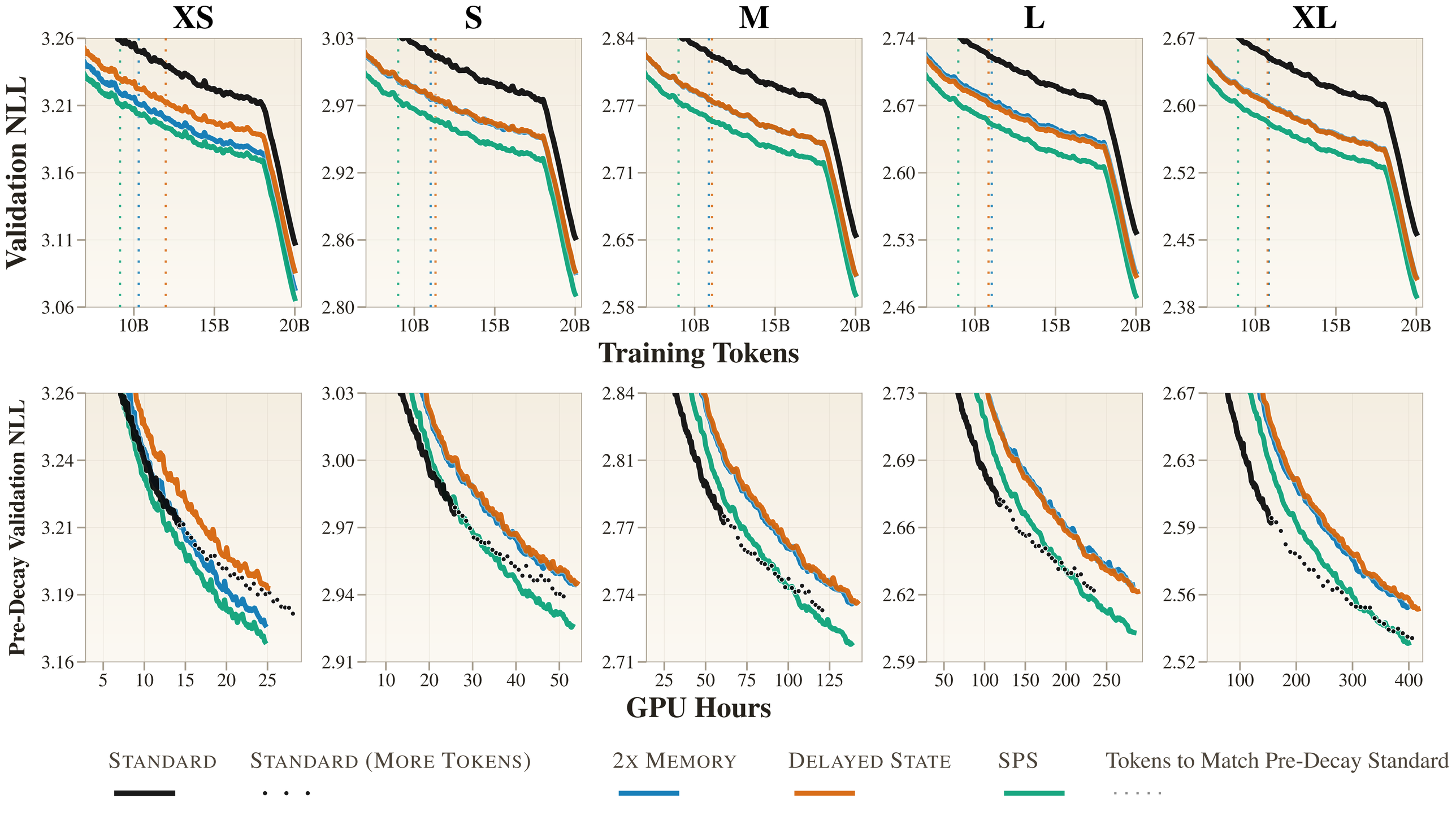

XS
S
M
L
XL
Validation NLL
3.26
3.21
3.16
3.11
3.06
3.03
2.97
2.92
2.86
2.80
2.84
2.77
2.71
2.65
2.58
2.74
2.67
2.60
2.53
2.46
2.67
2.60
2.52
2.45
2.38
10B
15B
20B
Training Tokens
Pre-Decay Validation NLL
3.26
3.24
3.21
3.19
3.16
3.03
3.00
2.97
2.94
2.91
2.84
2.81
2.77
2.74
2.71
2.73
2.69
2.66
2.62
2.59
2.67
2.63
2.59
2.56
2.52
5
10
15
20
25
10
20
30
40
50
25
50
75
100
125
50
100
150
200
250
100
200
300
400
GPU Hours
Standard
Standard (More Tokens)
2x Memory
Delayed State
SPS
Tokens to Match Pre-Decay Standard